Koo Foundation Sun Yat-Sen Cancer Center

Real-world clinical actionability of comprehensive genomic profiling in advanced solid tumors: a prospective single-center pan-cancer cohort

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Background

- Real-world actionability of comprehensive genomic profiling (CGP) in advanced solid tumors ranges 20–40% across published cohorts[1,2], mostly large US and European consortia.
- Asian single-center, first-pass implementation data are scarce.
- **Aim.** To characterise the biomarker landscape and therapy-match rate at a Taiwanese tertiary cancer center.

Methods

- **Cohort.** 62 consecutive late-stage solid-tumor patients, enrolled prospectively over 8 monthly batches.
- **Assay.** 523-gene panel (TruSight Oncology 500, DRAGEN v2.6): small variants, copy-number alterations, large rearrangements, tumor mutational burden (TMB), microsatellite instability (MSI) and genomic instability score (GIS; HRD proxy).
- **Annotation.** OncoKB[3] and CiVIC[4], tiered by AMP/ASCO/CAP[5].
- **Therapy match.** Stratified by OncoKB level, with resistance flags tracked (Table 1).

Term	Definition
On-label match	OncoKB Level 1, 2 or 3A
Off-label match	OncoKB Level 3B or 4
Resistance flag	OncoKB Level R1 or R2
TMB-High	≥10 mut/Mb
HRD-positive	GIS ≥42

Table 1. Definitions used throughout.

Cohort

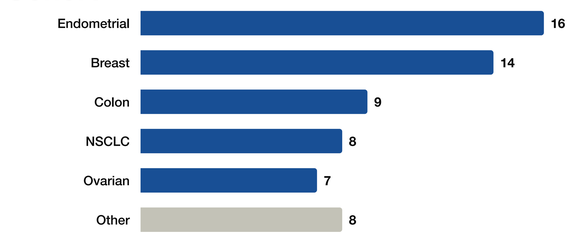


Figure 1. Patients by cancer type (n = 62; 11 types). Other: pancreatic (2); rectal, gastric, biliary, cervical, sarcoma and unspecified primary (1 each).

33.9% — **On-label match** — 21 of 62 · Level 1/2/3A

61.3% — **Off-label match** — 38 of 62 · Level 3B/4

12.9% — **TMB-High** — 8 of 62 · ≥10 mut/Mb

9.7% — **HRD-positive** — 6 of 62 · GIS ≥42

3.2% — **MSI-High** — 2 of 62

9.7% — **Resistance flag** — 6 of 62 · R1/R2

Results: genomic landscape

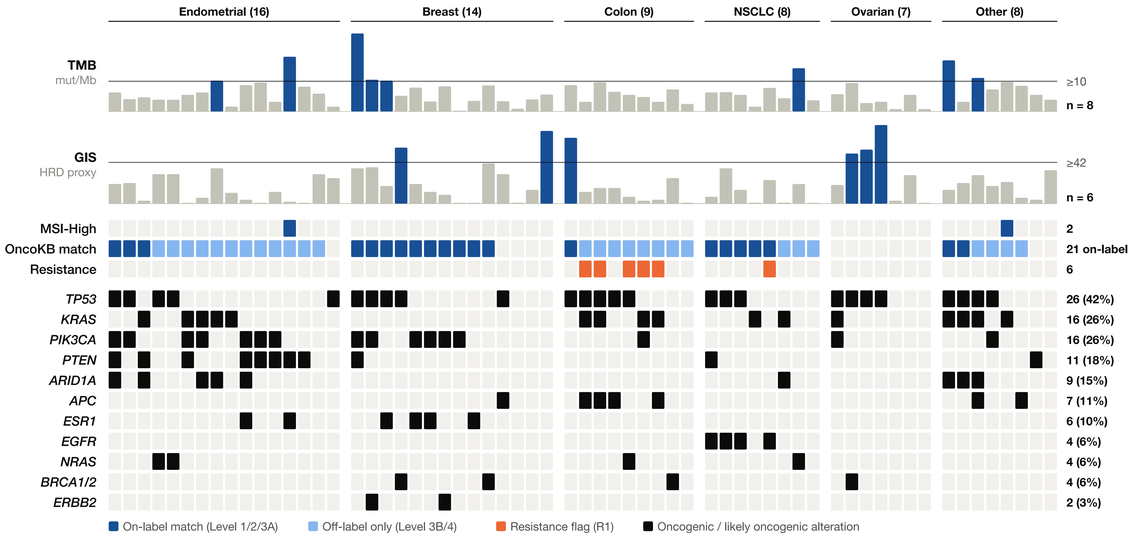


Figure 2. One column per patient, grouped by cancer type. Upper tracks: TMB and GIS (line: positivity threshold), MSI status, best OncoKB therapy level and resistance flags. Lower rows: oncogenic or likely oncogenic alterations in the most recurrent driver genes and in therapy-defining genes, with patient counts.

Therapy match by cancer type

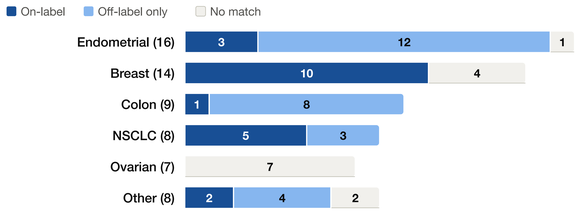


Figure 3. Best OncoKB therapy level per patient. Numbers are patients.

Biomarker-defined subsets

- **TMB-High, 8 (12.9%).** Breast 3, endometrial 2; pancreatic, NSCLC and cervical 1 each.
- **HRD-positive, 6 (9.7%).** Ovarian 3, breast 2, colon 1.
- **MSI-High, 2 (3.2%).** Endometrial 1, biliary 1.
- No ovarian tumor had an OncoKB-level match, yet 3 of 7 were HRD-positive.

On-label alterations

Gene	Patients	Cancer type	Level
PIK3CA	6	breast	1
EGFR	4	NSCLC	1
ESR1	4	breast	1
KRAS	3	pancreatic 2, NSCLC 1	3A
ERBB2	2	breast	1 / 2
PPP2R1A	2	endometrial	3A
BRAF	1	colon	1
BRCA1	1	breast	3A
BRCA2	1	breast	3A
FBXW7	1	endometrial	3A
PTEN	1	breast	1
TP53	1	NSCLC	3A

Table 2. Patients carrying an OncoKB Level 1/2/3A alteration, by gene.

Clinical highlights

- ***BRCA1* large rearrangement** (exon 7 loss, fold change 0.15) in an HRD-positive breast cancer, supporting on-label PARP inhibition.
- **Resistance flags in 6 patients.** *KRAS*- or *NRAS*-mutant colon cancer in 5 (anti-EGFR ineligibility) and *EGFR* T790M NSCLC in 1.
- **Recurrent drivers.** *TP53* (26 patients), *KRAS* (16), *PIK3CA* (16), *PTEN* (11), *ARID1A* (9), *APC* (7), *ESR1* (6).

Conclusions

- Routine pan-cancer CGP at a Taiwanese tertiary cancer center yielded an on-label therapy match in roughly 1 in 3 advanced solid tumors, consistent with international cohorts.
- Biomarker-defined subsets (TMB-High, HRD-positive, MSI-High) identified additional immunotherapy- or PARPi-eligible patients.
- Patient reports are openly browseable.

Limitations. Single center, 62 patients, no treatment-outcome data. HRD status is a panel-based GIS proxy.

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DISCLOSURE

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